

KNN Chapter - Exercise Answer Key

Exercise 1) Main idea behind KNN classification?

KNN predicts the class of a new sample by finding the K closest training samples and using a majority vote among their labels.

Exercise 2) Why is feature scaling important for KNN?

Because KNN uses distance. Large-scale features can dominate the distance calculation and distort neighbor selection. Scaling balances feature contributions.

Exercise 3) Distance between A(3,4) and B(7,1).

$dx = 4$; $dy = -3$; $d = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$.

Exercise 4) K=5 neighbors: A, A, B, A, B. Prediction?

A appears 3 times; B appears 2 times. Majority vote = A.

Exercise 5) Python code for KNN with K=5.

```
from sklearn.neighbors import KNeighborsClassifier
model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Exercise 6) Effect of changing K on accuracy.

Small K can overfit noise (high variance). Large K can oversmooth and underfit (high bias). Choose K using validation or cross-validation.